

Machine Learning:

Data Preprocessing

Xiuxia Du

Department of Bioinformatics and Genomics
College of Computing and Informatics
University of North Carolina at Charlotte

Introduction

- Extracting biological information from large omics datasets is a major challenge in bioinformatics.
 - genomics
 - transcriptomics
 - proteomics
 - metabolomics
 - exposomics

Factors that contribute to the challenge

- **Differences in orders of magnitude** between measured concentrations of biological entities. Entities in high concentrations are not necessarily more important than those present at low concentrations
 - Example: The average concentration of a signal molecule is much lower than the average concentration of a highly abundant compound like ATP.
- **Differences in the fold changes**
 - Example: The concentrations of metabolites in the central metabolism are generally relatively constant, while the concentrations of metabolites that are present in pathways of secondary metabolism usually show much larger differences in concentration depending on the environmental conditions.
- **Technical variation**
 - Batch effect, sample dilution, sample amount

Factors that contribute to the challenge

- **Measurement noise**
 - Chemical noise, random noise
- **High dimensionality and small sample size**
 - Hundreds to thousands of variables
 - Dozens or hundreds of samples
- **Heterogeneity**
 - Age, BMI, weight, height, weight to hip ratio
 - Cholesterol, A1C, glucose, LDL, HDL, ...

Purpose of Data Preprocessing

- Reduce technical artifacts
- Transform the data into a clean, consistent, and suitable format for statistical analysis and training machine learning models
- Improve the biological interpretability of the data

Data Preprocessing Tasks

- Audit data
- Handle missing values
- Normalize data
- Clean noise
- Center and scale variables
- Transform data
- Reduce dimensions
- Encode categorical variables
- Split the data for training, validation, and test

Data Centering

- Converts all the signal intensities to fluctuations around zero instead of around the mean of the signal.
- Adjust for differences in signal intensity
- Focus analysis attention on the fluctuating part of each signal by preserving only the variation

Data Scaling

- Divide each signal by a scaling factor that is different for each signal
- Methods
 - Autoscaling, also called unit variance scaling
 - Pareto scaling
 - Range scaling
 - Vast scaling
 - Level scaling

Data Scaling

Class	Method	Formula	Unit	Goal	Advantages	Disadvantages
I	Centering	$\tilde{x}_{ij} = x_{ij} - \bar{x}_i$	0	Focus on the differences and not the similarities in the data	Remove the offset from the data	When data is heteroscedastic, the effect of this pretreatment method is not always sufficient
II	Autoscaling	$\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_i}{s_i}$	(-)	Compare metabolites based on correlations	All metabolites become equally important	Inflation of the measurement errors
	Range scaling	$\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_i}{(x_{i_{\max}} - x_{i_{\min}})}$	(-)	Compare metabolites relative to the biological response range	All metabolites become equally important. Scaling is related to biology	Inflation of the measurement errors and sensitive to outliers
	Pareto scaling	$\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_i}{\sqrt{s_i}}$	0	Reduce the relative importance of large values, but keep data structure partially intact	Stays closer to the original measurement than autoscaling	Sensitive to large fold changes
	Vast scaling	$\tilde{x}_{ij} = \frac{(x_{ij} - \bar{x}_i)}{s_i} \cdot \frac{\bar{x}_i}{s_i}$	(-)	Focus on the metabolites that show small fluctuations	Aims for robustness, can use prior group knowledge	Not suited for large induced variation without group structure
	Level scaling	$\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_i}{\bar{x}_i}$	(-)	Focus on relative response	Suited for identification of e.g. biomarkers	Inflation of the measurement errors

Data Scaling: Auto vs Pareto

- Impact on data analysis
 - **Weighting of Variables:**
 - Auto scaling **equalizes all variables completely**, so even low-variance variables (like trace metabolites) can have equal influence as high-variance ones.
 - Pareto scaling **partially preserves variance structure**, so high-variance variables still carry more weight than low-variance ones.
- **Noise Sensitivity:**
 - Auto scaling can **inflate noise**: if a variable has very small variance (close to measurement noise), it will be scaled up disproportionately and might dominate models.
 - Pareto scaling is **less extreme**, so it's often considered a middle ground: balances the influence of small vs large variance variables.
- **Interpretability:**
 - Auto scaling results emphasize **relative changes**, not absolute abundance. Good for when all variables are equally important (e.g., exploratory PCA across heterogeneous features).
 - Pareto scaling tends to preserve **biological interpretability** better in omics data, since abundances still matter somewhat.

Data Scaling: Auto vs Pareto

- Practical consequences
 - **PCA/PLS results:**
 - With auto scaling: loadings will reflect equal weighting of all features. Variables with inherently small variation may appear as strong contributors.
 - With Pareto scaling: loadings will emphasize moderate-to-high variance features more, making results often **smoother and more biologically realistic**.
 - **Clustering:**
 - Auto scaling can make clusters form based on subtle differences in low-abundance features.
 - Pareto scaling tends to cluster more on **major metabolic/omic trends**.
 - **Feature selection:**
 - Auto scaling might highlight variables that were previously hidden by variance magnitude.
 - Pareto scaling prioritizes features with both strong signal and variance.

Data Scaling: Auto vs Pareto

- When to use each
 - **Auto scaling:**
 - When all variables are expected to be equally important.
 - When units differ strongly (e.g., combining glucose levels with enzyme activities).
 - For purely exploratory analysis.
 - **Pareto scaling:**
 - When you want to reduce dominance of large-variance variables but still respect magnitude.
 - Often the default in **metabolomics and proteomics**, since it reduces distortion of biological signal while still controlling variance.
- Rule of thumb
 - Use **auto scaling** if you want to “democratize” all features equally (risk: noise inflation).
 - Use **Pareto scaling** if you want a compromise that reduces variance dominance but maintains biological realism.

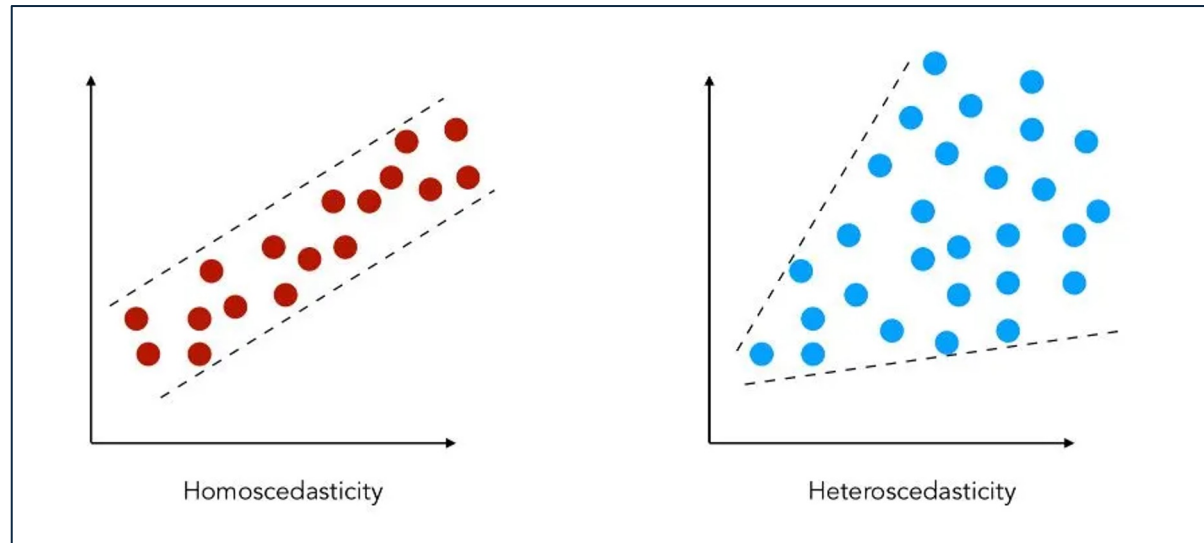
Data Transformation

- Methods
 - Log transformation
 - Power transformation

Class	Method	Formula	Unit	Goal	Advantages	Disadvantages
III	Log transformation	$\tilde{x}_{ij} = {}^{10}\log(x_{ij})$ $\hat{x}_{ij} = \tilde{x}_{ij} - \bar{\tilde{x}}_i$	Log O	Correct for heteroscedasticity, pseudo scaling. Make multiplicative models additive	Reduce heteroscedasticity, multiplicative effects become additive	Difficulties with values with large relative standard deviation and zeros
	Power transformation	$\tilde{x}_{ij} = \sqrt{x_{ij}}$ $\hat{x}_{ij} = \tilde{x}_{ij} - \bar{\tilde{x}}_i$	$\sqrt{O}$	Correct for heteroscedasticity, pseudo scaling	Reduce heteroscedasticity, no problems with small values	Choice for square root is arbitrary.

Data Transformation

- Correct for heteroscedasticity
 - Heteroscedasticity is a condition in regression analysis where the variance of the residual (error) terms is not constant across the range of the independent variables.
 - Instead of a consistent spread of data points around the regression line, heteroscedastic data shows a systematic change in the variance, often forming a cone or funnel shape.
 - This violates a key assumption of Ordinary Least Squares (OLS) regression, making the standard errors unreliable and the statistical inference misleading.



Normalization

- Purpose: Corrects for **systematic intensity shifts** caused by instrument sensitivity, injection amount, or sample loading differences.
- Transcriptomics (RNA-seq and gene expression)
 - **Counts per Million (CPM)**: Normalizes for sequencing depth by scaling read counts relative to total counts.
 - **RPKM/FPKM (Reads/Fragments Per Kilobase per Million)**: Normalizes for both sequencing depth and gene length.
 - **TPM (Transcripts Per Million)**: Similar to RPKM but more interpretable (values sum to $1e6$, making samples comparable).
 - **DESeq2 median ratio method**: Uses size factors estimated from the median ratio of observed counts to a reference.
 - **TMM (Trimmed Mean of M-values, edgeR)**: Adjusts for library size and composition bias.
 - **Quantile normalization**: Forces the distribution of expression levels to be the same across samples.

Normalization: Proteomics

- **Total Ion Current (TIC) normalization:** Scales intensities by the total ion count in each run.
- **Median normalization:** Centers intensities so that medians are aligned across samples.
- **Quantile normalization:** Makes intensity distributions identical across samples.
- **Variance Stabilization Normalization (VSN):** Applies transformation to stabilize variance across dynamic ranges.
- **Internal standards / spike-ins:** Normalization against known added proteins or peptides.

Normalization: Metabolomics

- **Internal standard normalization:** Dividing metabolite signals by stable isotope-labeled internal standards.
- **TIC normalization:** Scaling metabolite peak areas by total ion current (LC-MS).
- **Probabilistic Quotient Normalization (PQN):** Adjusts for dilution differences by scaling relative to a reference spectrum.
- **Median or sum normalization:** Scaling each sample's intensities to a common median or total.
- **Quantile normalization:** Aligns distributions across samples.
- **LOESS normalization (locally weighted regression):** Corrects intensity drifts (e.g., in time-course or batch runs).

Median Normalization: the Method

- **Input data:** You start with a protein intensity matrix (rows = proteins/peptides, columns = samples), often log-transformed to reduce heteroscedasticity.
- **Compute median per sample:**
 - For each sample/LC-MS run, calculate the **median intensity** across all proteins/peptides.
 - Example: Sample A median = 12.3, Sample B median = 11.5, Sample C median = 13.0.
- **Choose a reference median:** This could be:
 - The **global median** across all samples,
 - The **median of medians**, or
 - A specific reference sample.
- **Calculate scaling factors:**
 - For each sample, compute a **scaling factor** = (reference median $\div$ sample median).
 - Example: If the reference median = 12.3, then Sample B gets scaled by $12.3/11.5 \approx 1.07$.
- **Normalize intensities:**
 - Multiply all protein intensities in each sample by the corresponding scaling factor.
 - After this, all samples have approximately the **same median intensity**.

Median Normalization: the Method

- Assumes that **most proteins are not differentially expressed** (so their distribution centers should align).
- **Advantages**
 - Simple, robust, non-parametric — doesn't assume distribution shape beyond centering.
 - Reduces technical variation without overcorrecting.
- **Limitations**
 - If a large fraction of proteins truly change (e.g., strong biological effect, treatment vs control), median normalization may **mask real global shifts**.
 - Doesn't correct for variance heterogeneity (you might still apply log or variance-stabilizing transforms).

Quantile Normalization: the Method

Suppose you have an intensity matrix (rows = proteins/metabolites/genes, columns = samples).

- **Arrange each sample's values in order**
 - For each column (sample), sort intensities from smallest to largest.
 - Now you have a ranked list of values for each sample.
- **Compute the average (or median) across samples for each rank**
 - For rank 1 (smallest values), take the mean of all samples' smallest values.
 - For rank 2, take the mean of all samples' second smallest values, and so on.
 - This produces a "reference distribution" that all samples will share.
- **Replace values with the averaged quantiles**
 - For each sample, replace its *i*th ranked value with the *i*th average across samples.
 - Importantly: after replacement, **you restore the original ordering** of values within each sample.
- **Output**
 - Each sample now has exactly the **same distribution of intensities**.
 - The differences across samples are in the **relative ordering of proteins/metabolites/genes**, not in their overall distribution shape.

Quantile Normalization: the Method

- Effect
 - Eliminates systematic distributional differences between samples.
 - Strongly reduces technical variation from batch effects, loading, or instrument sensitivity.
 - Preserves **rank information** within each sample (the most abundant protein remains the most abundant).
- Advantages
 - Very effective when you believe biological differences should be reflected mainly in **relative ranks**, not overall distribution shape.
 - Widely used in microarrays, sometimes in proteomics/metabolomics before PCA or clustering.
- Limitations
 - Can **remove true biological global shifts** (e.g., if treatment causes widespread upregulation).
 - Assumes all samples should have **identical intensity distributions** — may be too strong if global biological effects are real.

Batch Effect Correction

- ComBat: linear modeling
- Limma: linear modeling
- SERRF: random forest
- MetNormalizer: support vector regression

References

- van den Berg RA, Hoefsloot HC, Westerhuis JA, Smilde AK, van der Werf MJ: Centering, scaling, and transformations: improving the biological information content of metabolomics data. BMC Genomics 2006, 7:142.